

西南交通大学 2021—2022 学年第(二) 学期考试试卷

课程代码 MECH007512 课程名称 材料力学 BII (补 A 卷) 考试时间 130 分钟

题号	一	二	三	四	五	六	七	八	总成绩
得分									

阅卷教师签字：_____

试卷共 8 个大题，请在答题纸上作答（不用抄写题目，但需注明题号）。

考试方式：线上考试

以下内容应在 130 分钟内完成。

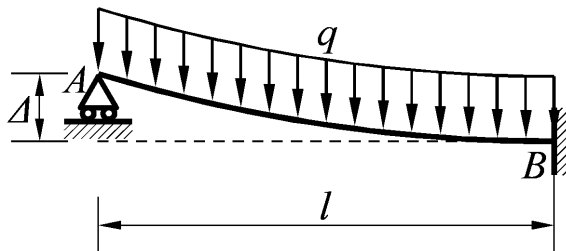
一、考试诚信承诺书。

我郑重承诺：我愿意服从学校本次考试的安排，承认考试成绩的有效性，并已经认真阅读、了解了《西南交通大学考试考场管理办法》和《西南交通大学本科生考试违规处理办法》，我愿意在本次考试过程中严格服从监考教师的相关指令安排，诚信考试。如果在考试过程中违反相关规定，我愿意接受《西南交通大学本科生考试违规处理办法》的规定处理。您是否同意：

A. 同意 B. 不同意

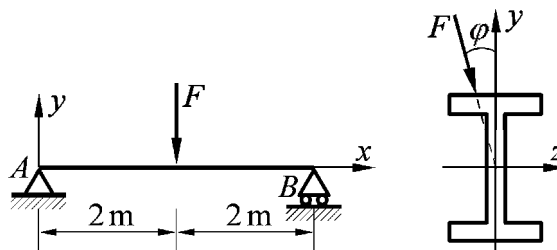
选择 B 选项，本次考试无效，记 0 分。

二、(15 分) 结构受力如图，已知梁的弯曲刚度为 EI ，受均布在 q 作用，从强度考虑为了使梁的受力最合理，试求支座 A 应上移的距离 Δ 。（略去剪切变形的影响）

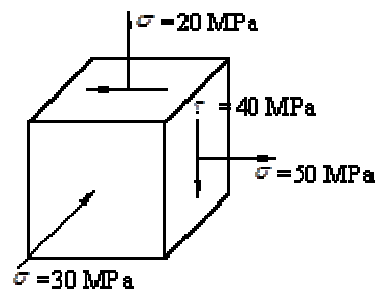


三、(15 分) 图示工字形截面的简支梁，弯曲截面系数 $W_z = 692\text{cm}^3$ ， $W_y = 70.8\text{cm}^3$ ，中间承受力 $F = 30\text{kN}$ ，

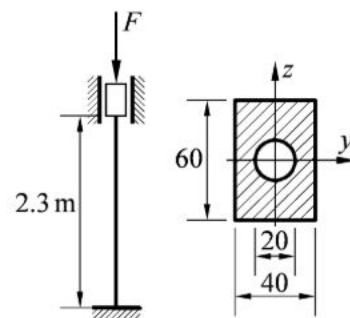
$\varphi = 15^\circ$ ， $[\sigma] = 170\text{MPa}$ 。试按正应力校核梁的强度。



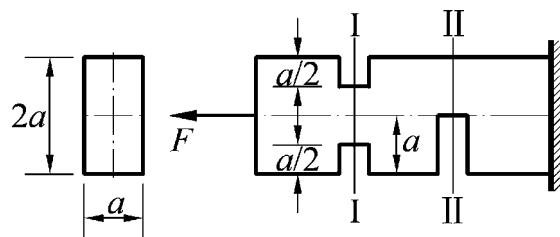
四、(20 分) 构件上某点处的应力状态如图所示。试求该点处的主应力及最大切应力。



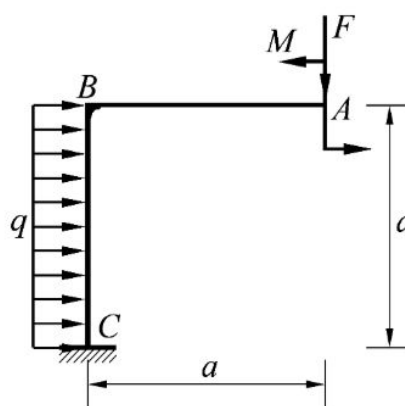
五、(15 分) 图示压杆， $F = 140\text{kN}$ ，当截面绕 z 轴失稳时，两端视为铰支；绕 y 轴失稳时，两端为固定端。已知：许用应力 $[\sigma] = 160\text{MPa}$ ，试校核压杆的稳定性。



六、(20 分) 矩形截面杆受轴向力 F 作用，在杆上开槽口如图所示。已知力 $F = 60\text{kN}$ ，长度 $a = 60\text{mm}$ 。试作截面 I-I 与 II-II 上的应力分布图。



七、(15 分) 平面刚架受载如图所示，已知： q ， a ， $F = qa$ ， $M = qa^2$ ，弯曲刚度为 EI ，试用能量法求截面 A 的转角和水平位移。



八：请按以下要求上传 2 张证明照片。

- (1) 请上传你自己、你已完成的答题纸（答题纸的签名页面）、你的身份证合照照片；
- (2) 请上传你的身份证和已完成的答题纸（答题纸的签名页面）的合照照片；

此题不答或不属实或不符合要求，本次考试无效，记 0 分。

答案:

1.

解: 求正弯矩最大点 $M(x) = F_{\text{支}}x - \frac{qx^2}{2}$, $\frac{dM(x)}{dx} = 0$ 得 $x_0 = \frac{F_{\text{支}}}{q}$

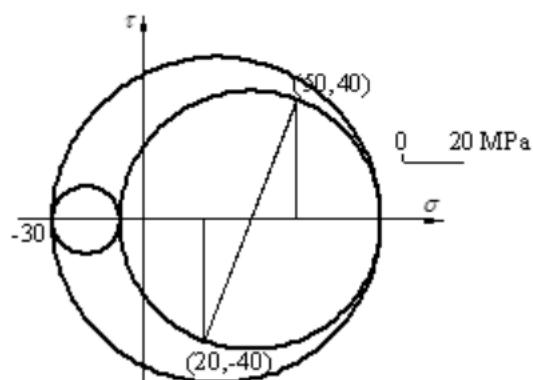
与B点负弯矩相等, $M(x_0) = -M_B$, $F_{\text{支}} = 0.415ql$, $\Delta = \frac{F_{\text{支}}l^3}{3EI} - \frac{ql^4}{8EI} = \frac{0.013ql^4}{EI}$

2.

解: 危险截面在中间处, $M_y = M_{\text{max}} \sin \varphi$, $M_x = M_{\text{max}} \cos \varphi$,

$\sigma_{\text{max}} = \frac{M_x}{W_x} + \frac{M_y}{W_y} = 152 \text{ MPa} < [\sigma]$, 满足强度条件。

3.



$$\sigma_1 = 77.7 \text{ MPa}$$

$$\sigma_2 = -7.7 \text{ MPa}$$

$$\sigma_3 = -30 \text{ MPa}$$

$$\tau_{\text{max}} = 53.9 \text{ MPa}$$

4.

解: $A = 2086 \text{ mm}^2$, $I_y = 31.2 \times 10^4 \text{ mm}^4$

$I_x = 71.2 \times 10^4 \text{ mm}^4$, $\lambda_y = 94$, $\lambda_x = 124.5 = \lambda_{\text{max}}$

$\varphi = 0.4368$, $\frac{F_N}{\varphi A} = 153.6 \text{ MPa} < [\sigma]$

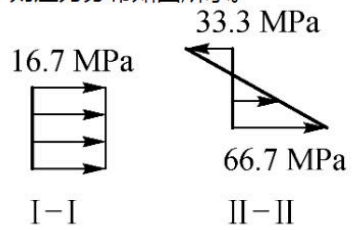
稳定

5.

解：截面 I-I , $\sigma_1 = \frac{F_{N1}}{A_1} = 16.7 \text{ MPa}$

截面 II-II , $\sigma_{\text{tmax}} = \frac{F_{N2}}{A_2} + \frac{M_2}{W} = 66.7 \text{ MPa}$ $\sigma_{\text{cmax}} = \frac{F_{N2}}{A_2} - \frac{M_2}{W} = -33.3 \text{ MPa}$

则应力分布如图所示。



6.

$$\theta_A = \frac{qa^3}{3EI} \text{ (逆时针) , } \delta_{AA} = \frac{qa^4}{8EI} \text{ (}\rightarrow\text{)}$$

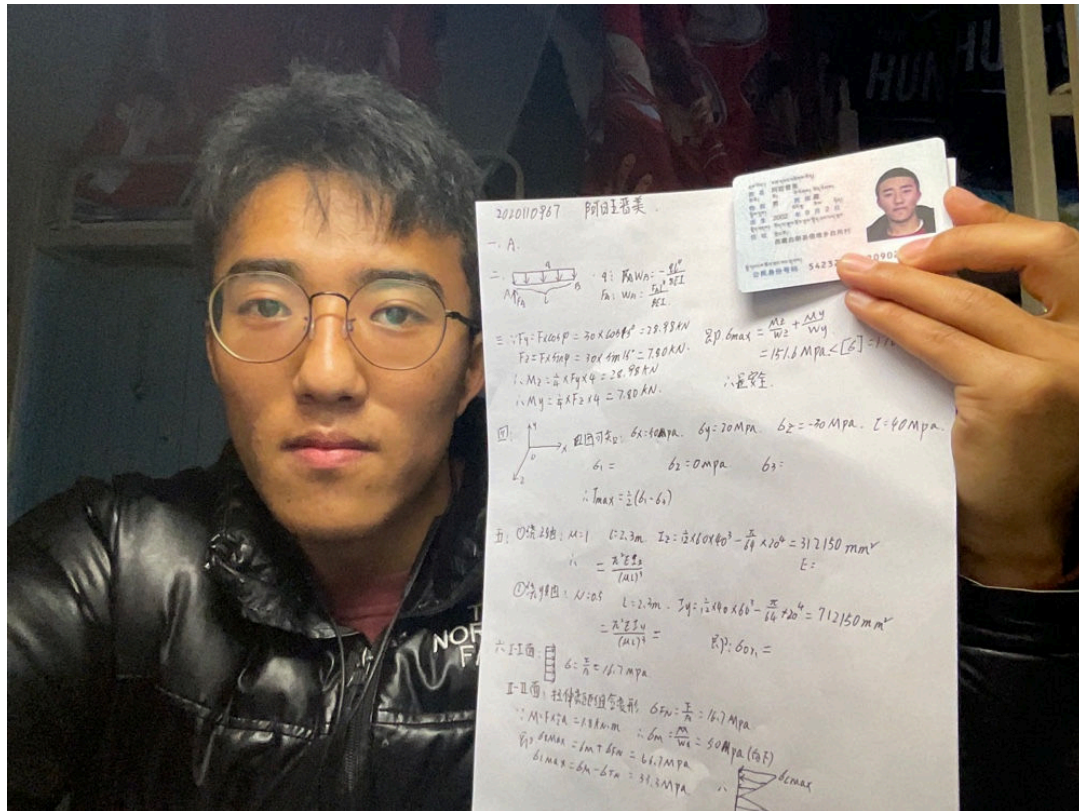
考场情况 （应到 11 人，缺考 3 人：范琦伟退考、孙健昆、罗远恒）

225	2020110940	朱新韬	土木工程学院	土木2020-22班	B2203	MECH007512	材料力学BII	1
226	2020110325	胡晨辉	土木工程学院	土木2020-01班	B2203	MECH007512	材料力学BII	1
227	2020110561	罗远恒	土木工程学院	土木2020-09班	B2205	MECH007512	材料力学BII	3
228	2020110582	孙嘉亮	土木工程学院	土木2020-10班	B2206	MECH007512	材料力学BII	4
229	2020110969	韩浩	土木工程学院	铁道2020-01班	B2206	MECH007512	材料力学BII	4
230	2020110680	扎西曲达	土木工程学院	土木2020-13班	B2207	MECH007512	材料力学BII	5
231	2020110790	王旭	土木工程学院	土木2020-17班	B2208	MECH007512	材料力学BII	6
232	2020110766	斯尔旦	土木工程学院	土木2020-16班	B2208	MECH007512	材料力学BII	6
233	2020110990	孙健昆	土木工程学院	铁道2020-02班	B2210	MECH007512	材料力学BII	8
234	2020110967	阿旺晋美	土木工程学院	铁道2020-01班	B2210	MECH007512	材料力学BII	8
235	2020110992	范琦伟	土木工程学院	铁道2020-02班	B2210	MECH007512	材料力学BII	8

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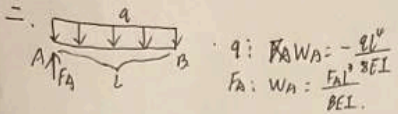
1.阿旺晋美



2020110967 阿日王晋美



一. A.



$$q: FAWA = -\frac{qL^4}{8EI}$$

$$FA: WA = \frac{FL^3}{8EI}$$

$$\therefore Fy = F \cos 15^\circ = 30 \times \cos 15^\circ = 28.98 \text{ kN}$$

$$Fz = F \sin 15^\circ = 30 \times \sin 15^\circ = 7.80 \text{ kN}$$

$$\therefore Mz = \frac{1}{4} \times Fy \times 4 = 28.98 \text{ kN}$$

$$\therefore My = \frac{1}{4} \times Fz \times 4 = 7.80 \text{ kN}$$

$$\sigma_{max} = \frac{Mz}{Wz} + \frac{My}{Wy}$$

$$= 151.6 \text{ MPa} < [\sigma] = 170 \text{ MPa}$$

∴ 是安全.

⑦: 由图可知: $\sigma_x = 50 \text{ MPa}$, $\sigma_y = 20 \text{ MPa}$, $\sigma_z = -30 \text{ MPa}$, $\tau = 40 \text{ MPa}$.

$$\sigma_1 = \sigma_2 = \sigma_3 =$$

$$\therefore \tau_{max} = \frac{1}{2}(\sigma_1 - \sigma_3)$$

五: ①绕z轴: $\mu = 1$, $L = 2.3 \text{ m}$, $I_z = \frac{1}{12} \times 60 \times 40^3 - \frac{\pi}{64} \times 20^4 = 312150 \text{ mm}^4$

$$\therefore \sigma = \frac{\pi^2 E I_z}{(\mu L)^3}$$

②绕y轴: $\mu = 0.5$, $L = 2.3 \text{ m}$, $I_y = \frac{1}{12} \times 40 \times 60^3 - \frac{\pi}{64} \times 20^4 = 712150 \text{ mm}^4$

$$= \frac{\pi^2 E I_y}{(\mu L)^3} =$$

⑧: $\sigma_{or} =$

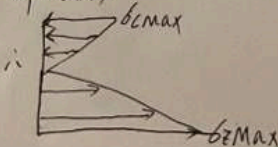
六 I-I面: $\sigma = \frac{F}{A} = 16.7 \text{ MPa}$

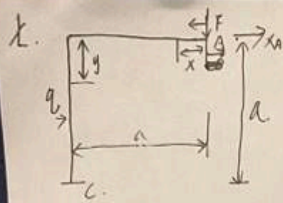
II-II面: 拉伸弯曲组合变形. $\sigma_{FN} = \frac{F}{A} = 16.7 \text{ MPa}$

$$\therefore M = F \times \frac{1}{2} L = 1.8 \text{ kN} \cdot \text{m} \quad \therefore \sigma_m = \frac{M}{W_z} = 50 \text{ MPa (向下)}$$

$$\sigma_{max} = \sigma_m + \sigma_{FN} = 66.7 \text{ MPa}$$

$$\sigma_{min} = \sigma_m - \sigma_{FN} = 33.3 \text{ MPa}$$





已知: $F = qa$, $M = qa^2$ 刚度 EI

① AB段: $M(x) = qa^2 - qa \cdot x$ 且 $\frac{dM(x)}{dx} = -q$

② BC段: $M(y) = qa^2 - qa^2 - \frac{1}{2} q y^2$ 且 $\frac{dM(y)}{dy} = -qy$

$\therefore \Delta A = \frac{1}{EI} \left[\int_0^a (qa^2 - qa \cdot x) dx + \int_0^a (-\frac{1}{2} q y^2) dy \right]$

$= \frac{qa^3}{3EI}$ (逆时针)

设在A处水平力为 X_A 如图: ① AB段: $M(x) = qa^2 - qa \cdot x$ 且 $\frac{dM(x)}{dx} = -q$

② BC段: $M(y) = qa^2 - qa^2 - \frac{1}{2} q y^2 - X_A \cdot y$ 且 $\frac{dM(y)}{dy} = -qy - X_A$

令 $X_A = 0$

$\therefore \Delta A_X = \frac{1}{EI} \left[\int_0^a 0 dx + \int_0^a (-\frac{1}{2} q y^2 + 0) dy \right]$

$= \frac{qa^4}{8EI}$ (水平向右)

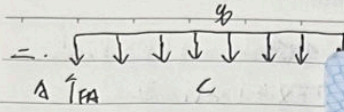


2. 斯尔旦



一同意。

2020110766 : 姓名 No. 日期



分布作用: $W_A = -\frac{ql^3}{8E}$

杆作用: $W_A = \frac{F_A l^3}{3E}$

令 $F_A = 0$

$\therefore \Delta = -W_A = \frac{ql^4}{8EI}$



$F_y = F_x \cos \varphi = 30 \times \cos 18^\circ = 28.98 \text{ kN}$

$F_z = F_x \sin \varphi = 30 \times \sin 18^\circ = 7.76 \text{ kN}$

$\therefore M_z = \frac{1}{4} \times F_y \times 4 = 28.98 \text{ kN} \cdot \text{m}$

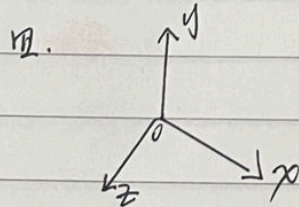
$M_y = \frac{1}{4} \times F_z \times 4 = 7.76 \text{ kN} \cdot \text{m}$

$\therefore \sigma_{\max} = \frac{M_z}{W_z} + \frac{M_y}{W_y}$

$= \frac{28.98 \times 10^3}{8.2 \times 10^{-6}} + \frac{7.76 \times 10^3}{70.8 \times 10^{-6}}$

$= 181.5 \text{ MPa} < [\sigma] = 170 \text{ MPa}$

$\therefore$ 安全

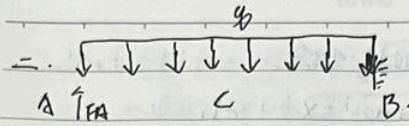


$\therefore \sigma_x = 80 \text{ MPa} \quad \sigma_y = 70 \text{ MPa}$

$\tau_{xy} = 60 \text{ MPa} \quad \sigma_z = -30 \text{ MPa}$

一、同意。

2020110766 · 姓名: 日期: No.:



$$q \text{ 作用: } W_A = -\frac{ql^4}{8EI}$$

$$F_A \text{ 作用: } W_A = \frac{F_A l^3}{3EI}$$

$$\sum F_A = 0$$

$$\therefore \Delta = -W_A = \frac{ql^4}{8EI}$$

$$\text{二、 } F_y = F_x \cos \varphi = 30 \times \cos 18^\circ = 28.98 \text{ kN}$$

$$F_z = F_x \sin \varphi = 30 \times \sin 18^\circ = 7.76 \text{ kN}$$

$$\therefore M_z = \frac{1}{4} \times F_y \times 4 = 28.98 \text{ kN} \cdot \text{m}$$

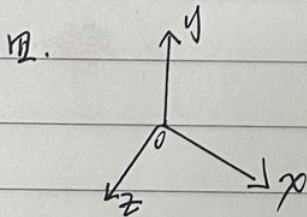
$$M_y = \frac{1}{4} \times F_z \times 4 = 7.76 \text{ kN} \cdot \text{m}$$

$$\therefore \sigma_{\max} = \frac{M_z}{W_z} + \frac{M_y}{W_y}$$

$$= \frac{28.98 \times 10^3}{892 \times 10^{-6}} + \frac{7.76 \times 10^3}{70.8 \times 10^{-6}}$$

$$= 141.5 \text{ MPa} < [\sigma] = 170 \text{ MPa}$$

$\therefore$ 合格



$$\therefore \sigma_x = 80 \text{ MPa} \quad \sigma_y = 70 \text{ MPa}$$

$$\tau_{xy} = 40 \text{ MPa} \quad \sigma_z = -30 \text{ MPa}$$

2020110766

斯尔里

Memo No.

Date

$$\sigma_1 = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$$

$$= \frac{1}{2} \times 70 + \frac{1}{2} \sqrt{30^2 + 4 \times 40^2}$$

$$= 77.72 \text{ MPa}$$

$$\therefore \sigma_1 = \frac{1}{2} \times 70 - \frac{1}{2} \sqrt{30^2 + 4 \times 40^2}$$

$$= -7.72 \text{ MPa}$$

$$\therefore \sigma_2 = 0 \text{ MPa}$$

$$\therefore J_{\max} = \frac{1}{2}(\sigma_1 - \sigma_3) = \frac{1}{2} \times (77.72 + 7.72)$$

$$= 42.72 \text{ MPa}$$

五. 绕z轴时. $m=1$ $l=2.3 \text{ m}$

$$I_z = \frac{1}{12} \times 60 \times 40^3 - \frac{\pi}{64} \times 20^4 = 312180 \text{ mm}^4$$

$$\therefore F_{N1} = \frac{\pi^2 EI_z}{(mL)^2}$$

$$= \frac{\pi^2 \times 200 \times 10^9 \times 312180 \times 10^{-12}}{(1 \times 2.3)^2}$$

$$= 116.4 \text{ kN}$$

绕y轴时. $m=0.5$ $l=2.3 \text{ m}$

$$I_y = \frac{1}{12} \times 40 \times 60^3 - \frac{\pi}{64} \times 20^4 = 712180 \text{ mm}^4$$

$$\therefore F_{N2} = \frac{\pi^2 EI_y}{(mL)^2}$$

$$= \frac{\pi^2 \times 200 \times 10^9 \times 712180 \times 10^{-12}}{(0.5 \times 2.3)^2}$$

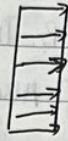
$$= 1061.85 \text{ kN}$$

$$\sigma_{\sigma 1} = \frac{F_{N1}}{A} = \frac{116.4 \times 10^3}{(40 \times 60 - \frac{\pi}{4} \times 20^2) \times 10^{-6}}$$

$$= 8 \text{ MPa} < [\sigma] = 160 \text{ MPa}$$

$\therefore$ 压杆稳定

六, I-I面



$$\sigma = \frac{F}{A} = \frac{60 \times 10^3}{0.06 \times 0.06} = 16.7 \text{ mpa}$$

II-II面.

拉伸弯曲组合变形.

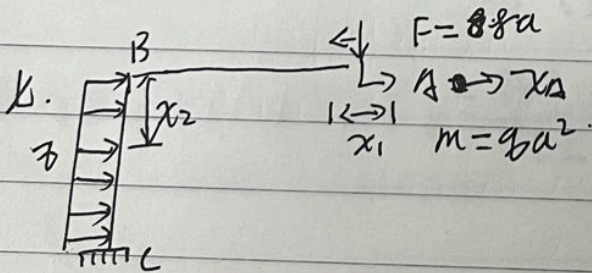
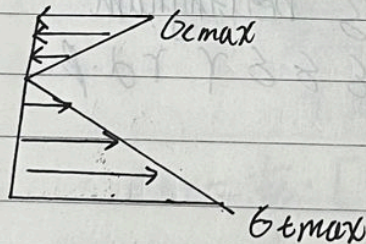
$$\therefore \sigma_{FN} = \frac{F}{A} = \frac{60 \times 10^3}{0.06 \times 0.06} = 16.7 \text{ mpa}$$

$$\therefore M = F \times \frac{1}{2}a = 60 \times \frac{1}{2} \times 0.06 = 1.8 \text{ kN} \cdot \text{m}$$

$$\therefore \sigma_m = \frac{M}{W_z} = \frac{1.8 \times 10^3}{\frac{1}{6} \times 0.06 \times 0.06^2} = \frac{1.8 \times 10^3}{\frac{1}{6} \times 0.06^3} = 80 \text{ mpa (下拉)}$$

$$\therefore \sigma_{tmax} = \sigma_m + \sigma_{FN} = 80 + 16.7 = 96.7 \text{ mpa}$$

$$\sigma_{cmax} = \sigma_m - \sigma_{FN} = 80 - 16.7 = 63.3 \text{ mpa}$$



AB段: $M(x_1) = qa^2 - qa \cdot x_1$

$$\frac{dM(x_1)}{dx_1} = -q$$

$$BC \text{ 段: } M(x_2) = qa^2 - qa \cdot a - \frac{1}{2} qx_2^2$$

$$\frac{dM(x_2)}{dx_2} = 1$$

$$\begin{aligned} \therefore \theta_A &= \frac{1}{EI} \cdot \left[\int_0^a (qa^2 - qax_1) dx_1 + \int_0^a (-\frac{1}{2} qx_2^2) dx_2 \right] \\ &= \frac{1}{EI} \cdot \left[qa^2 \cdot a - \frac{1}{2} qa \cdot a^2 + (-\frac{1}{6} qa^3) \right] \\ &= -\frac{qa^3}{3EI} (\nearrow) \end{aligned}$$

在 A 处假设水平力 X_A ($\rightarrow$)

$$\therefore AB \text{ 段: } M(x_1) = qa^2 - qa \cdot x_1$$

$$\frac{dM(x_1)}{dx_A} = 0$$

$$\therefore BC \text{ 段: } M(x_2) = qa^2 - qa \cdot a - \frac{1}{2} qx_2^2 - X_A x_2$$

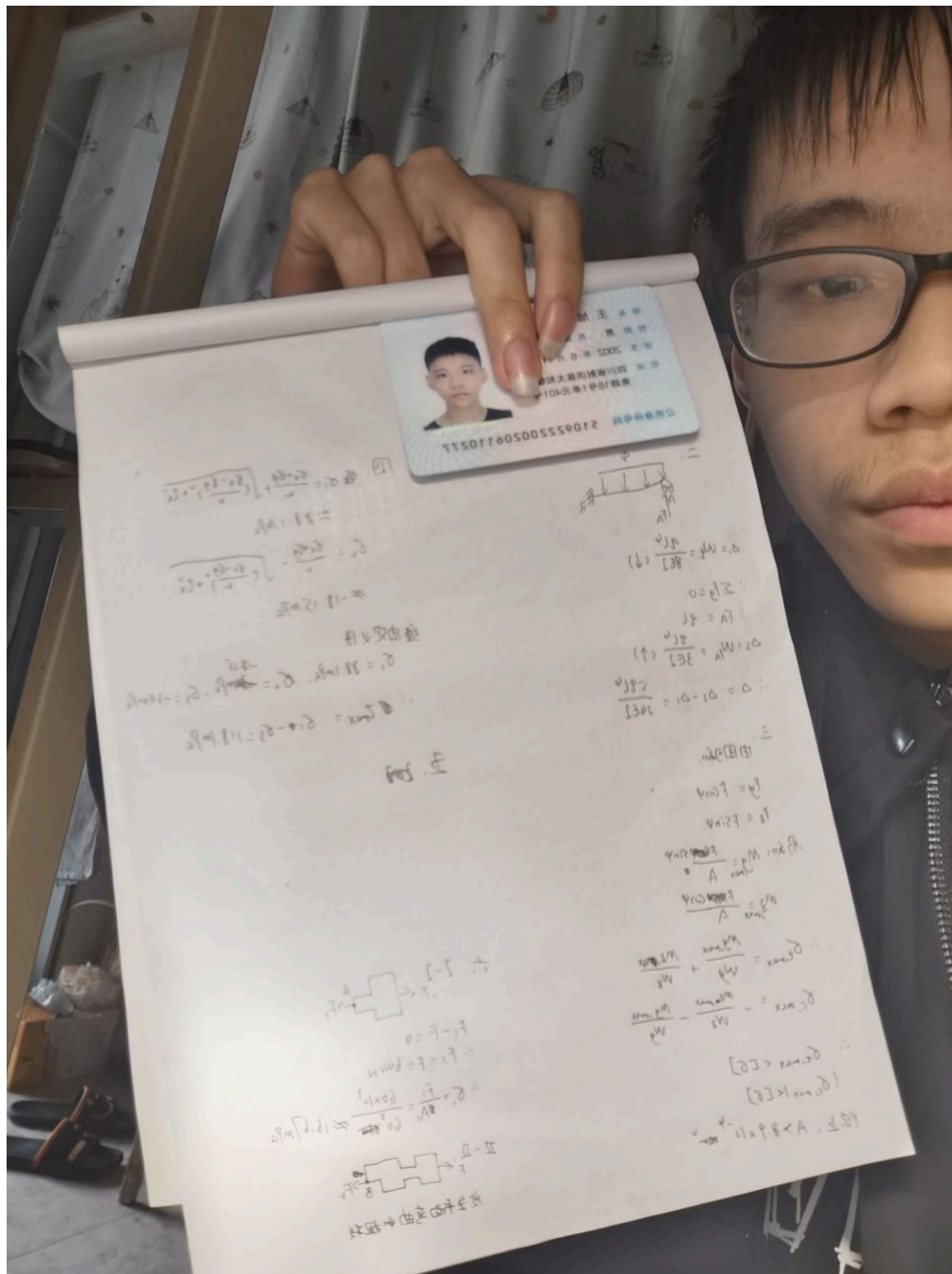
$$\frac{dM(x_2)}{dx_A} = -x_2$$

$$\text{令 } X_A = 0$$

$$\begin{aligned} \therefore \Delta_{AX} &= \frac{1}{EI} \cdot \left[\int_0^a 0 dx_1 + \int_0^a (\frac{1}{2} qx_2^3 + 0) dx_2 \right] \\ &= \frac{1}{EI} \cdot \left(\frac{1}{2} \times \frac{1}{4} \times qa^4 \right) \end{aligned}$$

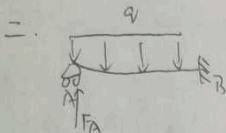
$$= \frac{qa^4}{8EI} (\rightarrow)$$

3.王旭



承(传)书

A



$$\Delta_1 = W_B = \frac{ql^4}{8EI} \quad (\downarrow)$$

$$\therefore \sum F_y = 0$$

$$\therefore F_A = ql$$

$$\Delta_2 = W_{FA} = \frac{ql^4}{3EI} \quad (\uparrow)$$

$$\therefore \Delta = \Delta_2 - \Delta_1 = \frac{5ql^4}{24EI}$$

三. 由图可知:

$$F_y = F \cos \varphi$$

$$F_z = F \sin \varphi$$

$$\text{易知: } M_{y, \max} = \frac{F \sin \varphi}{A}$$

$$M_{z, \max} = \frac{F \cos \varphi}{A}$$

$$\therefore \sigma_{t, \max} = \frac{M_{y, \max}}{W_y} + \frac{M_{z, \max}}{W_z}$$

$$\sigma_{c, \max} = -\frac{M_{z, \max}}{W_z} - \frac{M_{y, \max}}{W_y}$$

$$\therefore \sigma_{t, \max} < [\sigma]$$

$$|\sigma_{c, \max}| < [\sigma]$$

$$\text{综上, } A > 8.9 \times 10^{-4} \text{ m}^2$$

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\approx 88.1 \text{ MPa}$$

$$\sigma_2 = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\approx -18.15 \text{ MPa}$$

经由定义得

$$\sigma_1 = 88.1 \text{ MPa}, \sigma_2 = -18.15 \text{ MPa}, \sigma_3 = -50 \text{ MPa}$$

$$\therefore \sigma_{\max} = \sigma_1 - \sigma_3 = 138.1 \text{ MPa}$$

五. 乙

姓名 王旭

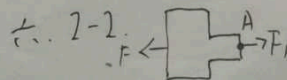
性别 男 民族 汉

出生 2002 年 6 月 11 日

住址 四川省射洪县太和镇环城南路18号1单元401号



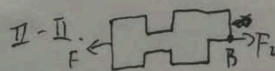
公民身份号码 510922200206110277



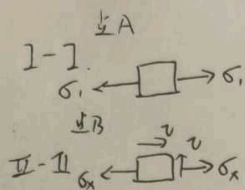
$$F_1 - F = 0$$

$$\therefore F_1 = F = 60 \text{ kN}$$

$$\therefore \sigma_1 = \frac{F_1}{A_1} = \frac{60 \times 10^3}{60^2} \approx 16.7 \text{ MPa}$$

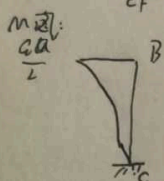


发生平面弯曲和扭转



t.

$$V_{EF} = \frac{M_F^2 R}{2E2}$$



$$\sum M_A = 0$$

$$M_F + \frac{qa'}{L} - qa'' = 0$$

$$\therefore M_F = \frac{qa''}{L}$$

$$\therefore V_{EF} = \frac{q^2 a^5}{8E2}$$

$$\Delta_1 = \frac{\partial V_{EF}}{\partial F} = \frac{qa^4}{4E2}$$

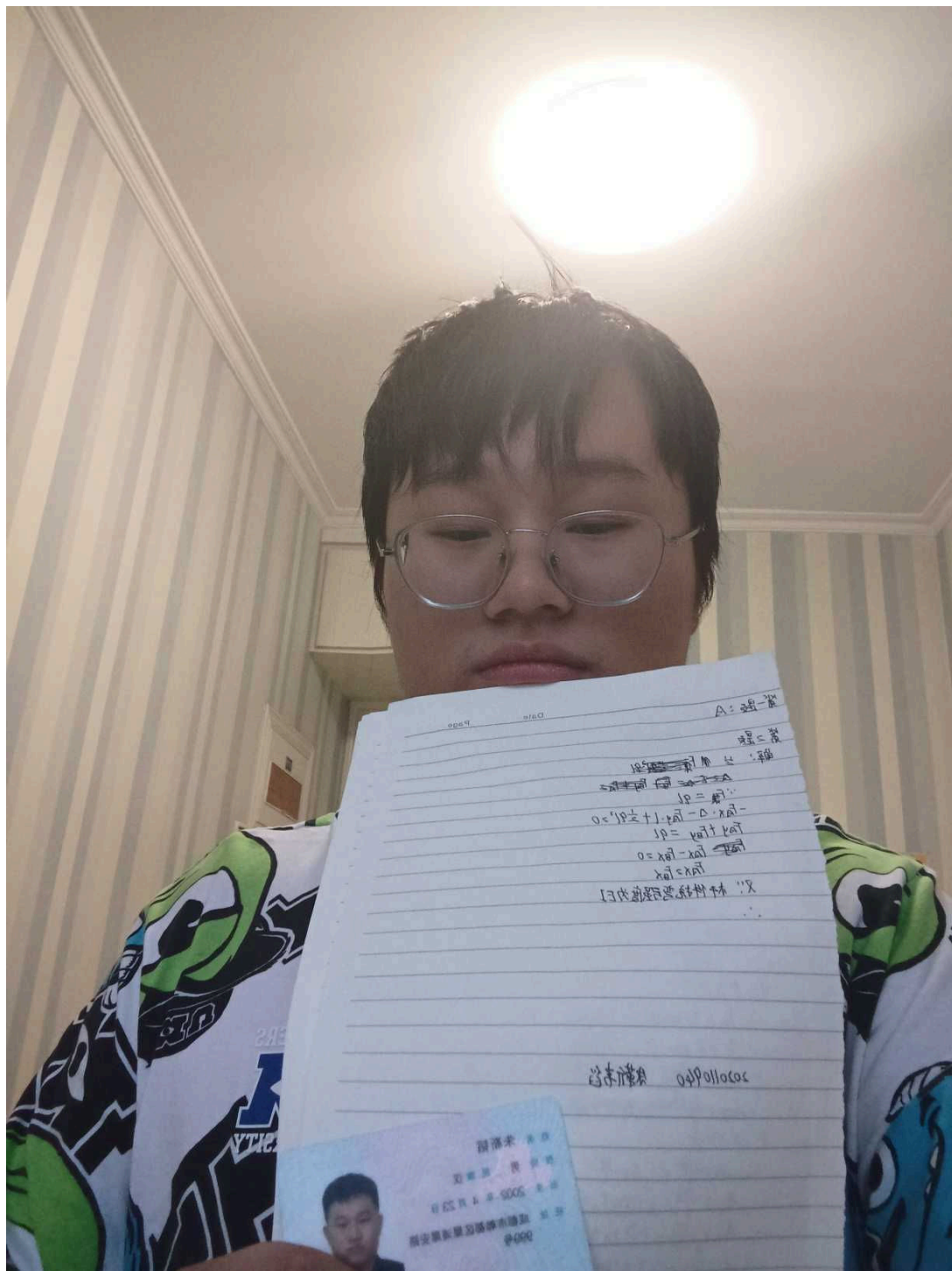
$$V_{Em} = \frac{M^2}{2E2} = \frac{q^2 a^4}{2E2}$$

$$\Delta_2 = \frac{\partial V_{Em}}{\partial F} = \frac{qa^3}{E2}$$

$$\therefore \Delta = \Delta_1 + \Delta_2 = \frac{q^2 a^4}{2E2} + \frac{qa^3}{E2}$$



4. 朱靳韬



第一题: A

Date

Page

第二题

解: 由 $\sum M = 0$

$$F_{Ax} \cdot l - F_{Ay} \cdot l + \frac{1}{2} q l^2 = 0$$

$$\therefore F_{Ax} = q l$$

$$-F_{Ax} \cdot l - F_{Ay} \cdot l + \frac{1}{2} q l^2 = 0$$

$$F_{Ay} + F_{By} = q l$$

$$F_{Ax} - F_{By} = 0$$

$$F_{Ax} = F_{By}$$

又: 杆件抗弯强度为 EI

2020110940 朱靳轺

姓名 朱靳轺

性别 男 民族 汉

出生 2002 年 4 月 23 日

住址 成都市郫都区犀浦犀安路
999号



公民身份号码 652923200204231410

第三题

Date.

Page.

解: $F_{Ax} + F_{Ay} = 0$

$$F_{Ax} + F_{Ay} = 0$$

$$\therefore F_{Ax} = 0$$

$$\therefore F_{Ax} + F_{Ay} = 0$$

$$F_{Ay} + F_{By} = 30 \text{ kN}$$

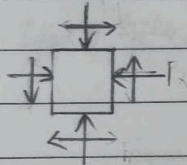
对 A 点取矩

$$F_{By} \cdot 4 - F \cdot 2 = 0$$

$$F_{By} = \frac{F}{2} = 15 \text{ kN}$$

$$\therefore F_{Ay} = 15 \text{ kN}$$

在 I 字梁上取一单元体



$$\therefore \theta = 15^\circ$$

$$\therefore \sigma_{\max} = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\therefore F_y = F \cdot \cos 15^\circ = 28.98 \text{ kN}$$

$$\therefore \sigma_{\max} = 3.615 \text{ MPa}$$

$$F_x = F \cdot \sin 15^\circ = 7.76 \text{ kN}$$

$$\text{同理 } \sigma_{\min} = -2079.321 \text{ MPa}$$

$$\therefore \sigma_{\max} = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\therefore \sigma_{\max} < [\sigma] = 170 \text{ MPa}$$

$$\sigma_{\min} = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

符合第一强度理论

$$\therefore \sigma = \frac{F}{A}$$

$$\therefore \sigma_x = \frac{F_x}{A} \quad \therefore A = \frac{W}{l}$$

$$\therefore A_y = \frac{W_y}{l} = 1.77 \times 10^{-5} \text{ m}^2$$

$$A_z = \frac{W_z}{l} = 1.73 \times 10^{-4} \text{ m}^2$$

$$\therefore \sigma_x = \frac{F_x}{A_y} = -438.418 \text{ MPa}$$

$$\sigma_y = \frac{F_y}{A_z} = -1637.288 \text{ MPa}$$

$$\tau_x = \frac{F_y}{A_z} = 86.705 \text{ MPa}$$

例四

Date.

Page.

$$\begin{aligned}\text{解 } \sigma_{\max} &= \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \\ &= \frac{50 + 20}{2} + \sqrt{(15)^2 + (-40)^2} \\ &= 35 + 42.72 \\ &= 77.72 \text{ MPa}\end{aligned}$$

$$\begin{aligned}\text{同理: } \sigma_{\min} &= \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \\ &= 35 - 42.72 \\ &= -7.72 \text{ MPa}\end{aligned}$$

$$\therefore \sigma_1 = 77.72 \text{ MPa}$$

$$\sigma_2 = 0 \text{ MPa}$$

$$\sigma_3 = -7.72 \text{ MPa}$$

$$\therefore \tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = 42.72 \text{ MPa}$$

五

解: $I_z = \frac{60 \times 40^3}{12}$

$$\lambda_z = \frac{\mu_z \cdot l}{i_z} = \frac{1 \times l}{i_z} = \frac{l}{i_z}$$

$$\lambda_y = \frac{\mu_y \cdot l}{i_y} = \frac{0.5 \times l}{i_y} = \frac{l}{2i_y}$$

$$\lambda_z = \lambda_y \Rightarrow \frac{l}{i_z} = \frac{l}{2i_y} \Rightarrow i_z = 2i_y$$

$$i_z = \sqrt{\frac{I_z}{A}}$$

$$\therefore A = 60 \times 40 - \pi \times \left(\frac{d}{2}\right)^2$$

$$= 2085.84 \text{ mm}^2 \approx 2.08 \times 10^3 \text{ mm}^2$$

$$I_y = \frac{40 \times 60^3}{12} - \frac{\pi \times (20)^4}{32}$$

$$I_y = \frac{40 \times 60^3}{12} - \frac{\pi \times (20)^4}{32}$$

$$= 7.04 \times 10^5 \text{ mm}^4$$

$$I_z = \frac{60 \times 40^3}{12} - \frac{\pi \times (20)^4}{32}$$

$$= 3.04 \times 10^5 \text{ mm}^4$$

$$\therefore i_y = \sqrt{\frac{I_y}{A}} = \sqrt{\frac{7.04 \times 10^5}{2.08 \times 10^3}} = 18.4 \text{ mm}$$

$$i_z = \sqrt{\frac{I_z}{A}} = \sqrt{\frac{3.04 \times 10^5}{2.08 \times 10^3}} = 12.1 \text{ mm}$$

$$\therefore \lambda_y = \frac{l}{i_y} = \frac{2.3 \times 10^3}{18.4} = 125$$

$$\lambda_z = \frac{l}{i_z} = \frac{2.3 \times 10^3}{12.1} = 190.1$$

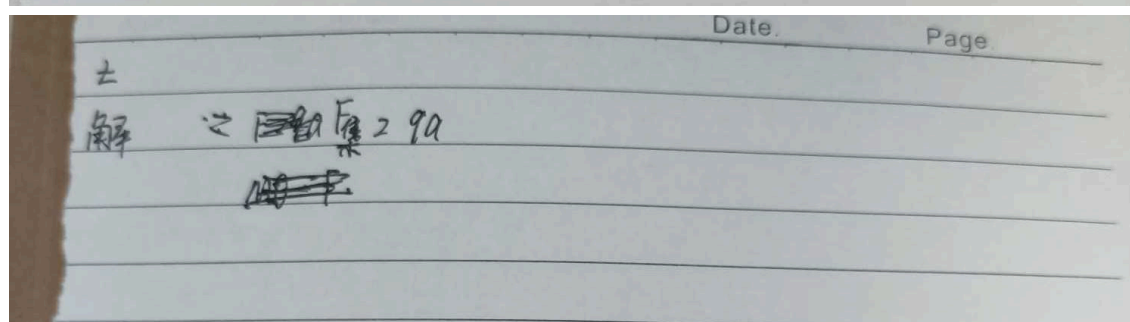
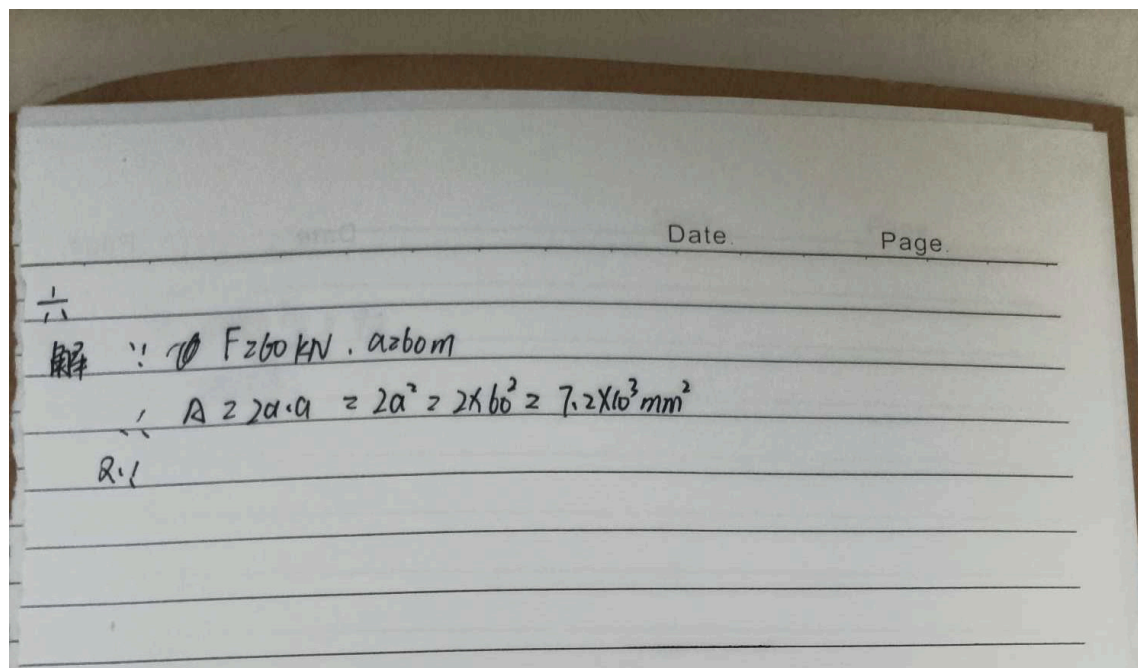
$$\therefore \lambda_z > \lambda_y$$

$$\therefore \sigma_{cr} = \frac{\pi^2 E}{\lambda_z^2} = \frac{\pi^2 \times 200 \times 10^3}{190.1^2}$$

$$\sigma_{cr} = \frac{P}{A} = \frac{140 \text{ kN}}{2.08 \times 10^3} = 67.3 \text{ MPa}$$

$$\therefore \sigma_{cr} < [\sigma] = 160 \text{ MPa}$$

符合



5.扎西曲达

2020110680 扎西曲达

1. 已知:

2. 求: 求 F_n 和 q .

3. 解:

由 $F_n = 20$

$0 = -WA = \frac{ql^4}{8EI}$

$WA = \frac{ql^4}{8EI}$

$F_y = F \cos \varphi = 30 \times \cos 15^\circ = 28.98 \text{ kN}$

$F_x = F \sin \varphi = 30 \times \sin 15^\circ = 7.76 \text{ MPa}$

$M_x = F_y \cdot x = 28.98 \text{ kN}$

$M_y = 7.76 \text{ kN}$

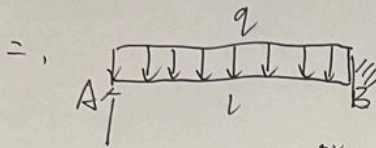
Diagram 1: A beam of length l with a uniformly distributed load q acting downwards. The beam is supported at the left end by a pin support and at the right end by a roller support.

Diagram 2: A beam of length l with a point load F acting downwards at the right end. The beam is supported at the left end by a pin support. The horizontal distance from the support to the point load is x .

Diagram 3: A cross-section of a beam with a vertical force F_y and a horizontal force F_x acting on it. The vertical force F_y is acting downwards and the horizontal force F_x is acting to the right.

2020110680 扎西曲达

1. 同意.



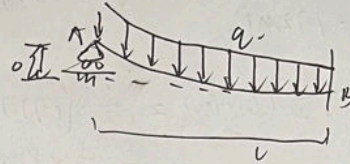
$$W_A = -\frac{ql^4}{8EI}$$

$$W_A = \frac{F_A l^3}{3EI}$$

令 $F_A = 0$.

$$0 = -W_A = \frac{ql^4}{8EI}$$

弯矩图 M . 均布 q .

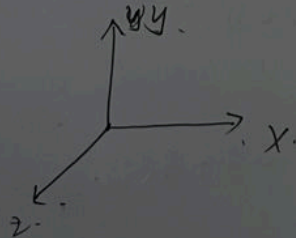
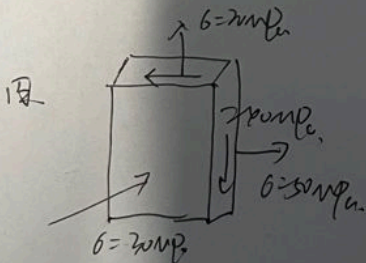
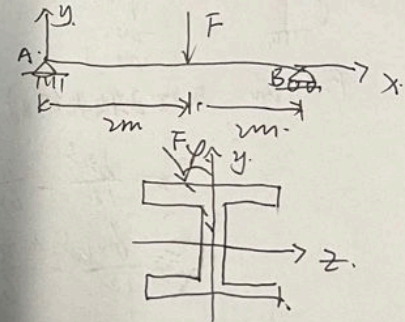


$$F_y = F \cos \varphi = 30 \times \cos 15^\circ = 28.98 \text{ kN}$$

$$F_z = F \sin \varphi = 30 \times \sin 15^\circ = 7.76 \text{ kN}$$

$$M_z = F_y \cdot l = 28.98 \text{ kN}$$

$$M_y = 7.76 \text{ kN}$$



$$\sigma_x = 50 \text{ MPa}$$

$$\sigma_y = 20 \text{ MPa}, \sigma_z = 30 \text{ MPa}$$

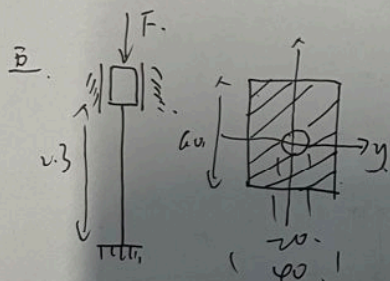
$$\tau_{xy} = 40 \text{ MPa}$$

主应力 σ_{max} .

$$\begin{aligned} \sigma_1 &= \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2} \\ &= \frac{1}{2} \times 70 + \frac{1}{2} \sqrt{(50 - 20)^2 + 4 \times 40^2} \\ &= 35 + \frac{1}{2} \sqrt{30^2 + 4 \times 1600} = 77.72 \text{ MPa} \end{aligned}$$

$$\begin{aligned}\sigma_z &= \frac{1}{2}(\sigma_x + \sigma_y) - \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2} \\ &= \frac{1}{2}(50 + 20) - \frac{1}{2}\sqrt{(50 - 20)^2 + 4 \times 80^2} \\ &= -7.72 \text{ MPa}\end{aligned}$$

$$\begin{aligned}\tau_{\max} &= \frac{1}{2}(\sigma_1 - \sigma_3) = \frac{1}{2} \times [177.72 - (-7.72)] \\ &= \frac{1}{2} \times 185.44 = 92.72 \text{ MPa}\end{aligned}$$



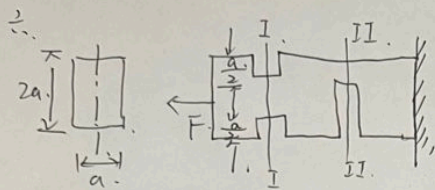
$$\tau_{67} = 160 \text{ MPa}$$

$F = 140 \text{ kN}$. 且绕 z 轴旋转时, 内力为常数,

$$\begin{aligned}I_z &= \frac{bh^3}{12} - \frac{\pi d^4}{64} \\ &= \frac{60 \times 100^3}{12} - \frac{\pi \times 20^4}{64} = 312150 \text{ mm}^4\end{aligned}$$

$$\therefore F_{cr1} = \frac{\pi^2 EI_z}{(nL)^2} = \frac{\pi^2 \times E I_z}{(n \times 2)^2} =$$

$F = 140 \text{ kN}$. 且绕 y 轴旋转时, 内力为常数.



$F = 60 \text{ kN}$, $a = 60 \text{ mm}$.

II-面



$\sigma = \frac{F}{A} = 16.7 \text{ MPa}$

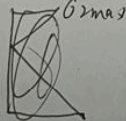
II-面

$\sigma_{FV} = \frac{F}{A} = 16.7 \text{ MPa}$

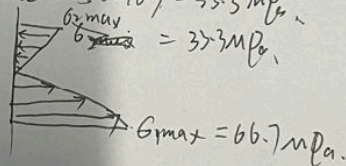
$M = \frac{1}{2} a F = \frac{1}{2} \times 0.06 \times 60 = 1.8 \text{ kNm}$

$\sigma_n = \frac{M}{W_z} = \frac{1.8}{\frac{1}{6} \times 0.06^3} = 50 \text{ MPa}$

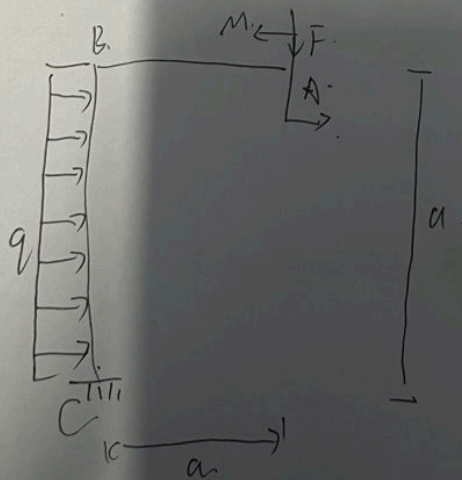
$\sigma_{max} = 50 + 16.7 = 66.7 \text{ MPa}$



$\sigma_{max} = 50 - 16.7 = 33.3 \text{ MPa}$



7. 题



$F = qa$, $M = qa^2$ EI.

AB: $M(x_1) = qa^2 - qa \cdot x_1$
 $\frac{dM}{dx} = -1$ III.

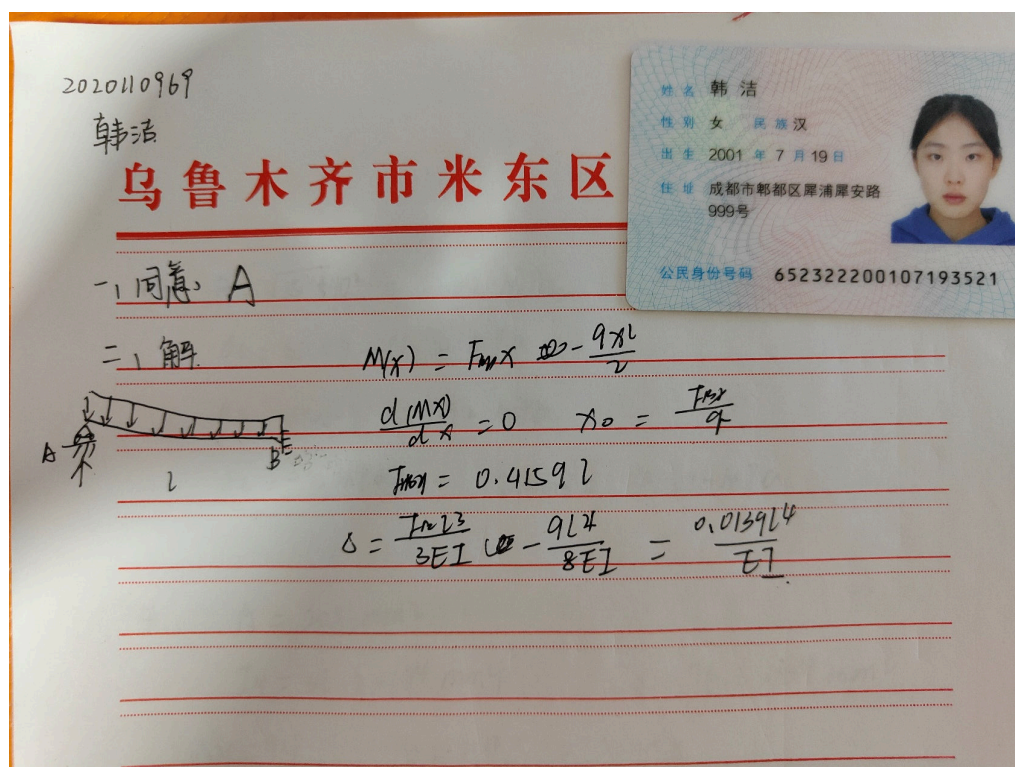
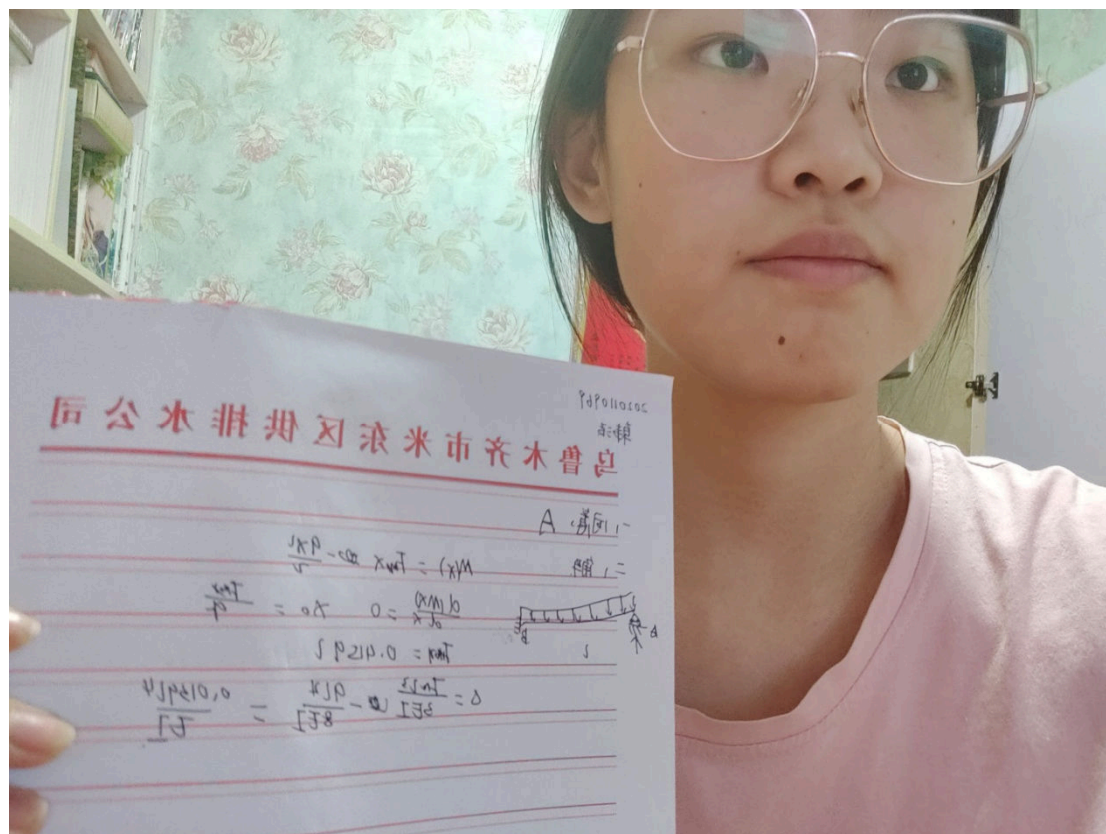
BC: $M(x_2) = qa^2 - qa \cdot a - \frac{1}{2} qx_2^2$
 $= qa^2 - qa^2 - \frac{1}{2} qx_2^2$
 $= -\frac{1}{2} qx_2^2$

$\frac{dM_{min}}{dx} = 0$

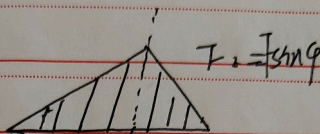
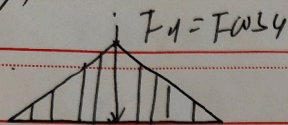
高刚度 = EI

6.韩浩





三、



$$\begin{aligned} \sigma_{max} &= \frac{M_y \max}{W_y} + \frac{M_z \max}{W_z} = \frac{FL}{4} \left(\frac{\sin 4}{W_y} + \frac{\cos 4}{W_z} \right) \\ &= \frac{30 \times 10^3 \times 4}{4} \times \left(\frac{\sin 15^\circ}{70.8 \times 10^6} + \frac{\cos 15^\circ}{692 \times 10^6} \right) \\ &= 152 \times 10^6 \text{ Pa} = 152 \text{ MPa} < [\sigma] \quad \text{满足} \end{aligned}$$

10. X (50, 40) Y (20, 40) z (30, 0)

$$y = \sqrt{\left(\frac{50-20}{2}\right)^2 + 40^2} = 42.72$$

圆心坐标为 (35, 0)

$$\sigma_1 = 42.72 + 35 = 77.72 \text{ MPa}$$

$$\sigma_3 = 35 - 42.72 = -7.72 \text{ MPa}$$

$$\sigma_2 = 35 \text{ MPa}$$

$$\tau_{max} = \frac{\sigma_1 - \sigma_3}{2} = 42.72 \text{ MPa}$$

五, $A = 2086 \text{ mm}^2$.

$I_y = 31.2 \times 10^4 \text{ mm}^4$

$I_z = 71.2 \times 10^4 \text{ mm}^4$

$\lambda_y = 94$

$\lambda_z = 124.5 = \lambda_{max}$

$\varphi = 0.4368$

$\frac{N}{\varphi A} = 153.6 \text{ MPa} < [67]$

该压杆稳定

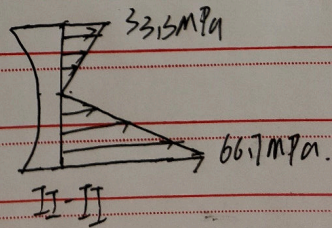
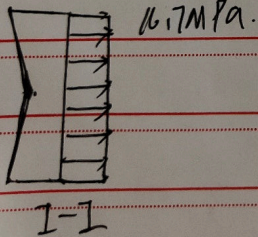
六, 截面 I-I

截面 II-II 中

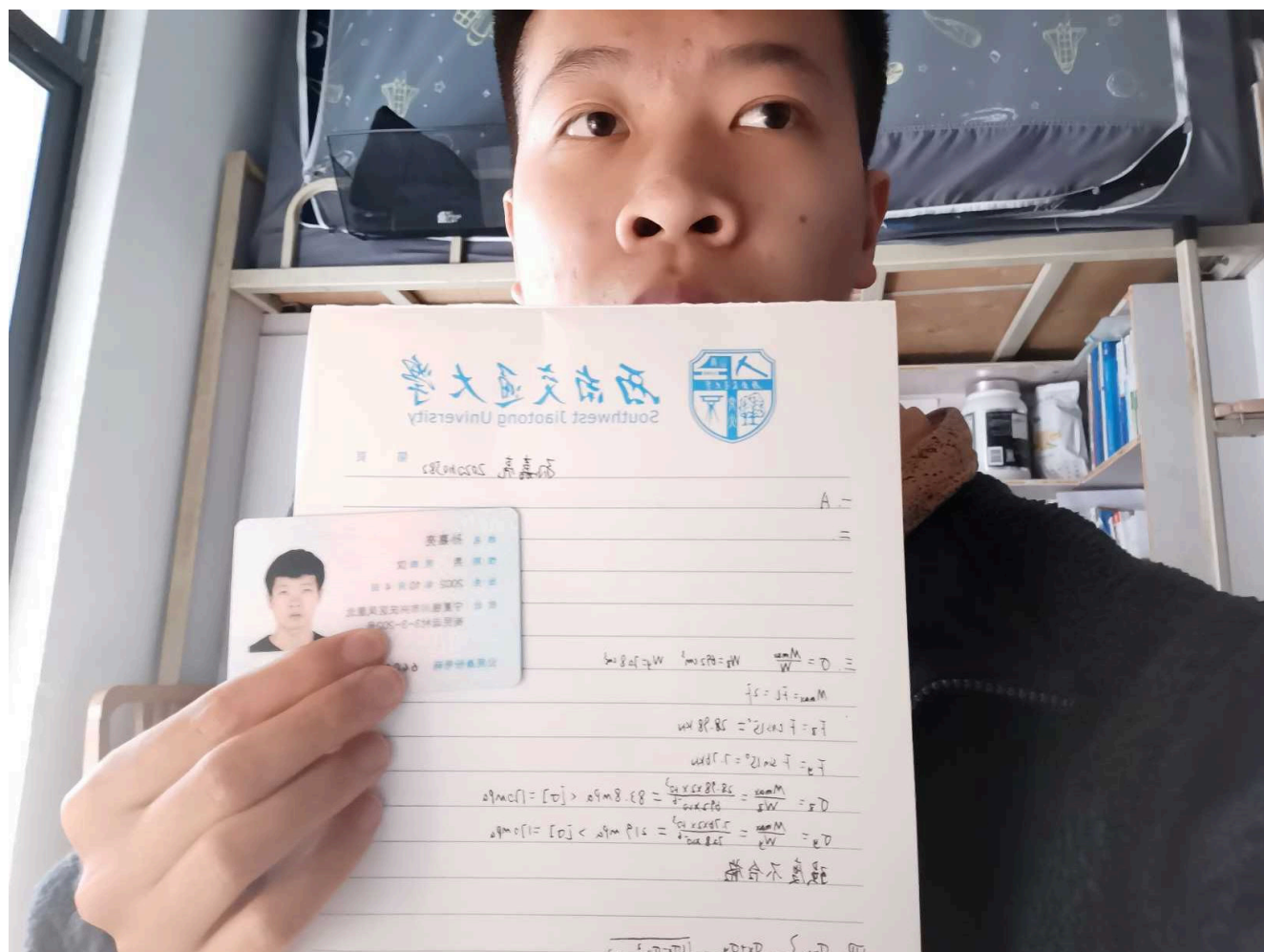
$\sigma_1 = \frac{F_{N1}}{A} = 16.7 \text{ MPa}$

$(\sigma_x)_{max} = \frac{F_{N2}}{A_x} + \frac{M_2}{W} = 66.7 \text{ MPa}$

$(\sigma_x)_{min} = \frac{F_{N2}}{A} - \frac{M_2}{W} = -33.3 \text{ MPa}$



7.孙嘉亮





西南交通大学
Southwest Jiaotong University

孙嘉亮 202010582 第 页

一. A

二.

姓名 孙嘉亮

性别 男 民族 汉

出生 2002 年 10 月 4 日

住址 宁夏银川市兴庆区凤凰北
街民运村3-3-202号



公民身份号码 640122200210040035

$$\text{三. } \sigma = \frac{M_{\max}}{W} \quad W_z = 692 \text{ cm}^3 \quad W_y = 728 \text{ cm}^3$$

$$M_{\max} = FL = 2F$$

$$F_z = F \cos 15^\circ = 28.98 \text{ kN}$$

$$F_y = F \sin 15^\circ = 7.76 \text{ kN}$$

$$\sigma_z = \frac{M_{\max}}{W_z} = \frac{28.98 \times 2 \times 10^3}{692 \times 10^{-6}} = 83.8 \text{ MPa} < [\sigma] = 120 \text{ MPa}$$

$$\sigma_y = \frac{M_{\max}}{W_y} = \frac{7.76 \times 2 \times 10^3}{728 \times 10^{-6}} = 219 \text{ MPa} > [\sigma] = 120 \text{ MPa}$$

强度不合格.

$$\text{四. } \left. \begin{array}{l} \sigma_{\max} \\ \sigma_{\min} \end{array} \right\} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_x^2}$$

$$\sigma_y = 20 \text{ MPa} \quad \tau_x = 40 \text{ MPa} \quad \sigma_x = -60 \text{ MPa}$$

$$\therefore \sigma_{\max} = \frac{-60 + 20}{2} + \sqrt{\left(\frac{-60 - 20}{2}\right)^2 + 40^2} = 36.57 \text{ MPa} = \sigma_1 \quad \sigma_2 = 0$$

$$\sigma_{\min} = \frac{-60 + 20}{2} - \sqrt{\left(\frac{-60 - 20}{2}\right)^2 + 40^2} = -76.57 \text{ MPa} = \sigma_3$$

$$\therefore \tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = \frac{36.57 - (-76.57)}{2} = 56.57 \text{ MPa}$$

犀浦校区地址: 四川省成都市高新区西部园区西南交通大学 邮编: 611756
峨眉校区地址: 四川省峨眉山市西南交通大学 邮编: 614202

九里校区地址: 四川省成都市二环路北一段111号 邮编: 610031



五: 双轴失稳 $\mu=1.0$ y 轴失稳 $\mu=0.5$

$$I_x = \frac{bh^3}{12} - \frac{1}{64}\pi D^4 = \frac{40 \times 60^3}{12} - \frac{1}{64}\pi \cdot 20^4 = 712146.02$$

$$A = 40 \times 60 - \pi \left(\frac{D}{2}\right)^2 = 2086 \text{ mm}^2$$

$$I_y = \frac{hb^3}{12} - \frac{1}{64}\pi D^4 = \frac{60 \times 40^3}{12} - \frac{1}{64}\pi \cdot 20^4 = 312146.02$$

$$i = \sqrt{\frac{I_{\min}}{A}} = 12.23 \text{ mm}$$

$$\eta = \frac{\mu l}{i} = \frac{0.5 \times 2.3}{12.23 \times 10^{-3}} = 94.03$$

$$F_{cr} = \sigma_{cr} A = \frac{\pi^2 E}{\eta^2} A$$

$$\sigma = \frac{F}{A} \quad F = 140 \text{ kN} \quad [\sigma] = 160 \text{ MPa}$$

11.





7. $\theta_A = \theta_1 + \theta_2 + \theta_3$ 顺时针为正

$$= -\frac{M_A}{EI} + \frac{qa^3}{6EI} + \frac{Fa^3}{3EI}$$

$$F = qa \quad M = qa^2$$

$$\therefore \theta_A = -\frac{qa^3}{EI} + \frac{qa^3}{6EI} + \frac{qa^3}{3EI}$$

$$= -\frac{qa^3}{2EI} \quad (7)$$

$$M: \theta_1 = \frac{M_A}{EI} \quad y = \frac{M_A^2}{2EI}$$

$$qa: \theta_2 = \frac{qa^3}{6EI} \quad y = \frac{qa^4}{8EI}$$

$$F: \theta_3 = \frac{Fa^3}{3EI} \quad y = \frac{FL^3}{3EI}$$

$$W_A = W_1 + \cancel{W_2} + \cancel{W_3}$$

$$= \frac{qa^3}{2EI}$$

姓名 孙嘉亮

性别 男 民族 汉

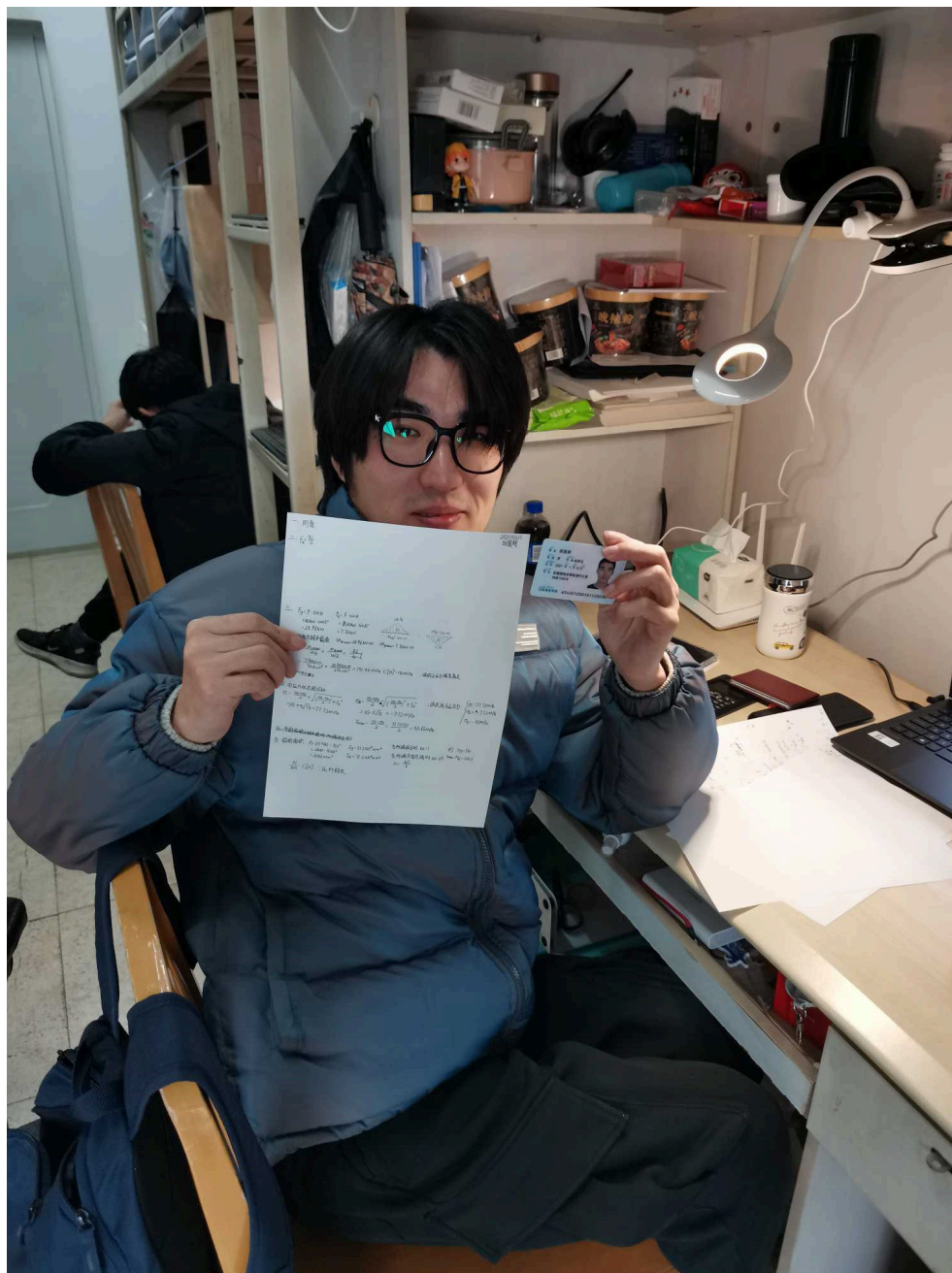
出生 2002 年 10 月 4 日

住址 宁夏银川市兴庆区凤凰北
街民运村3-3-202号



公民身份号码 640122200210040035

8.胡晨晖



一、同意

二、 $F_A = \frac{9L}{2}$

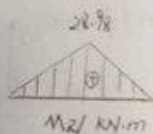
12001873, A^A Pn

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胡晨辉

姓名 胡晨辉
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三、 $F_y = F \cdot \cos 15^\circ$ $F_z = F \cdot \sin 15^\circ$
 $= 30 \text{ kN} \cdot \cos 15^\circ = 28.98 \text{ kN}$
 $= 30 \text{ kN} \cdot \sin 15^\circ = 7.76 \text{ kN}$



危险截面为跨中截面 $M_{z \max} = 28.98 \text{ kN} \cdot \text{m}$ $M_{y \max} = 7.76 \text{ kN} \cdot \text{m}$

$\sigma_{t, \max} = \frac{M_{y \max}}{W_y} + \frac{M_{z \max}}{W_z} = \frac{FL}{4}$

$= \frac{7.76 \text{ kN} \cdot \text{m}}{70.8 \text{ cm}^3} + \frac{28.98 \text{ kN} \cdot \text{m}}{6920 \text{ cm}^3} = 151.48 \text{ MPa} < [\sigma] = 170 \text{ MPa}$ $\therefore$ 梁的正应力强度满足。

$= 151.48$

四、由应力状态图可知：

$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$
 $= 35 + 5\sqrt{3} = 77.72 \text{ MPa}$

$\sigma_2 = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$
 $= 35 - 5\sqrt{3} = -7.72 \text{ MPa}$

$\therefore$ 该点处主应力：
 $\begin{cases} \sigma_1 = 77.72 \text{ MPa} \\ \sigma_2 = -7.72 \text{ MPa} \\ \sigma_3 = -30 \text{ MPa} \end{cases}$

$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = \frac{77.72 + 30}{2} = 53.86 \text{ MPa}$

五、当截面绕z轴失稳时，两端铰支时

五、截面面积： $A = 60 \times 40 - \pi r^2$ $I_y = 31.2 \times 10^4 \text{ mm}^4$
 $= 2400 - \pi \times 10^2$ $I_z = 71.2 \times 10^4 \text{ mm}^4$
 $= 2086 \text{ mm}^2$

当两端铰支时， $\mu = 1$

则 $\lambda_y = 94$

当两端为固定端时， $\mu = 0.5$ $\lambda_{\text{max}} = 78 = 124.5$

$\lambda = \frac{\mu L}{i}$

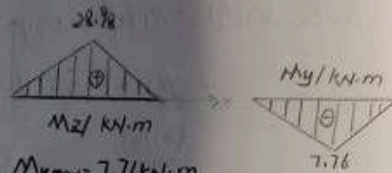
$\frac{N}{\phi A} < [\sigma] \therefore$ 压杆稳定。

一、同意

2020110325
胡晨辉

二、 $F_A = \frac{9L}{2}$

三、 $F_y = F \cdot \cos 4^\circ$ $F_z = F \cdot \sin 4^\circ$
 $= 30 \text{ kN} \cdot \cos 15^\circ$ $= 30 \text{ kN} \cdot \sin 15^\circ$
 $= 28.98 \text{ kN}$ $= 7.76 \text{ kN}$



危险截面为跨中截面 $M_{z \max} = 28.98 \text{ kN} \cdot \text{m}$ $M_{y \max} = 7.76 \text{ kN} \cdot \text{m}$

$$\sigma_{t, \max} = \frac{M_{y \max}}{W_y} + \frac{M_{z \max}}{W_z} = \frac{FL}{4}$$

$$= \frac{7.76 \text{ kN} \cdot \text{m}}{70.8 \text{ cm}^3} + \frac{28.98 \text{ kN} \cdot \text{m}}{692 \text{ cm}^3} = 151.48 \text{ MPa} < [\sigma] = 170 \text{ MPa} \therefore \text{梁的正应力强度满足}$$

$$= 151.48$$

四、由应力状态图可知:

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= 35 + 5\sqrt{3} = 77.72 \text{ MPa}$$

$$\sigma_2 = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= 35 - 5\sqrt{3} = -7.72 \text{ MPa}$$

$$\therefore \text{该点处主应力: } \begin{cases} \sigma_1 = 77.72 \text{ MPa} \\ \sigma_2 = -7.72 \text{ MPa} \\ \sigma_3 = -30 \text{ MPa} \end{cases}$$

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = \frac{77.72 + 30}{2} = 53.86 \text{ MPa}$$

五、当截面绕y轴失稳时, 两端铰支时:

五、截面面积: $A = 60 \times 40 - \pi r^2$ $I_y = 31.2 \times 10^4 \text{ mm}^4$
 $= 2400 - \pi \times 10^2$ $I_z = 71.2 \times 10^4 \text{ mm}^4$
 $= 2086 \text{ mm}^2$

当两端铰支时, $\mu = 1$

$$\lambda = \frac{l}{i_y} = 94$$

当两端为固定端时, $\mu = 0.5$ $\lambda_{\max} = \lambda_z = 124.5$

$$\lambda = \frac{\mu l}{i}$$

$$\frac{N}{\phi A} < [\sigma] \therefore \text{压杆稳定}$$

七. 设水平虚力为 $F_{\text{虚}}$.

AB段有 $M(x) = -M + F \cdot x$

$$\frac{\partial M(x)}{\partial F_{\text{虚}}} = 0$$

$$\frac{\partial M(x)}{\partial M} = -1$$

BC段: $M(y) = -M + F \cdot a + qy \cdot \frac{y}{2} + F_{\text{虚}} \cdot y$

$$= -M + F \cdot a + \frac{q}{2} y^2 + F_{\text{虚}} \cdot y$$

$$\frac{\partial M(y)}{\partial F_{\text{虚}}} = y \quad \frac{\partial M(y)}{\partial M} = -1$$

$$\Delta x = \frac{1}{EI} \int_0^a M(x) \cdot \frac{\partial M(x)}{\partial F_{\text{虚}}} dx + \frac{1}{EI} \int_0^l M(y) \cdot \frac{\partial M(y)}{\partial F_{\text{虚}}} dy$$

$$= \frac{1}{EI} \int_0^l (-M + Fx) \cdot 0 \cdot dx + \frac{1}{EI} \int_0^l (-M + Fa + \frac{q}{2} y^2 + F_{\text{虚}} y) \cdot y \cdot dy$$

$$= \frac{A^5}{8EI}$$